



I have created these robots to help me with my work. Robots are specifically designed to do different tasks. Let us see how robots are programmed to operate.

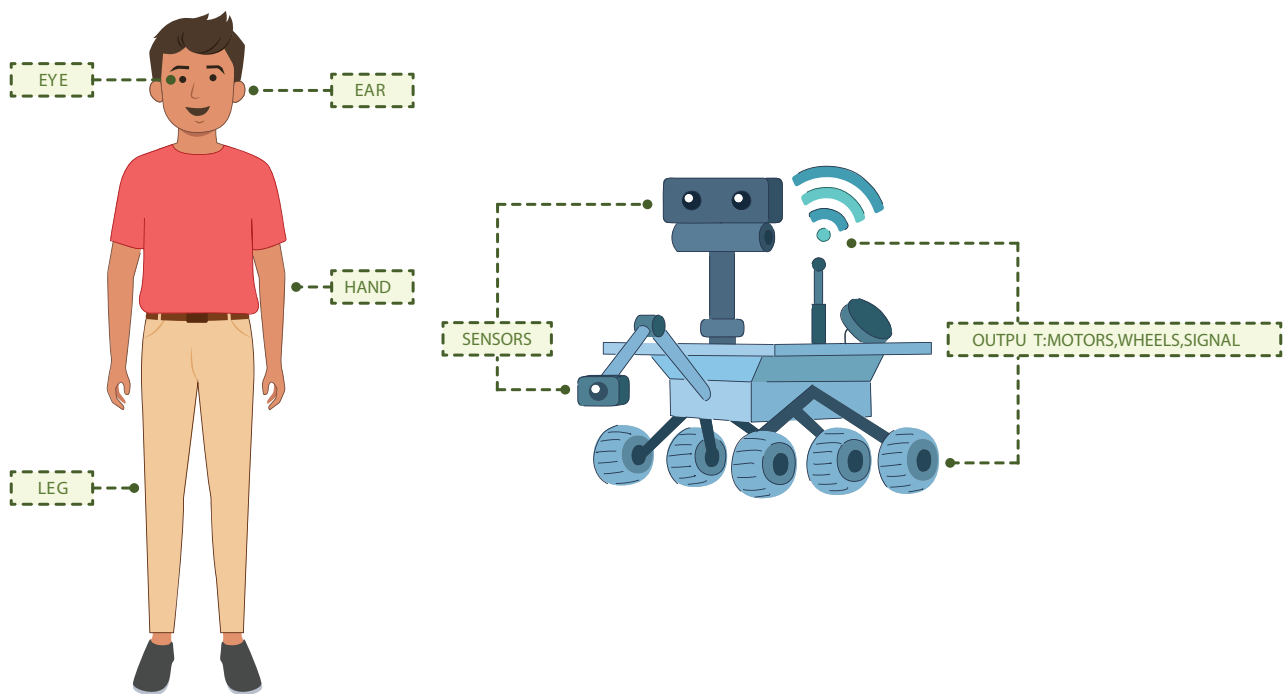
Note:

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What is a robot?

A robot is a machine that is programmed with the help of a computer. It is capable of automatically carrying out a complex series of actions. A robot may be guided by an external control device or have the control embedded within it. A robot can have any form. It may look like a human, or an animal, or a machine like a car, truck, etc. A robot is designed to perform desired tasks. Robots can be autonomous – self driven or semi-autonomous - human operated.

A living being collects information through its senses (eyes, ears, nose, skin, tongue). It processes information in its brain, and then acts using its limbs (legs, hands). A robot collects information from its surroundings (senses). It analyses that data (thinks) to perform a dedicated task based on the analysis (acts).



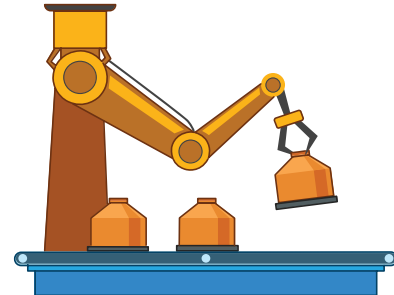
A robot senses information through inputs from sensors or a controller. It analyses the information using a processor and acts using actuators (e.g. motors) as an output. It requires a power source (like batteries) to get input, process it and provide output.

Application of robots

A few applications of robots are:

- **Manufacturing Industries**

A robot's work is divided into three parts: material handling, processing operations, and assembly and inspection.



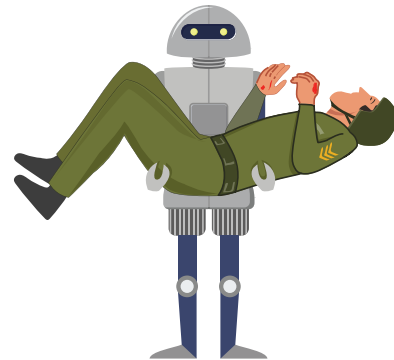
- **Medical Industry**

Robots are used to assist doctors in surgeries. They are also used for lab automation and medical device packaging.



- **Military**

In the military, robots tend to the injured army personnel.



- **Research Fields**

Robots are important to conduct research in unknown territories. They are used under water and on other planets.



Since 1979, robots and robotics have evolved rapidly and become an almost invisible presence in our lives. We have used robots in every field, from manufacturing and the military, to the space program. But the most common application that we are familiar with is a robotic vehicle. Have you played with a toy car or plane which is controlled by a remote? These are examples of a robotic vehicle.

What is a robotic vehicle?

A robotic vehicle is a special type of machine designed to move on the ground, in the air, under water and in space without a human on board. It is controlled and operated by sensor-based control systems.

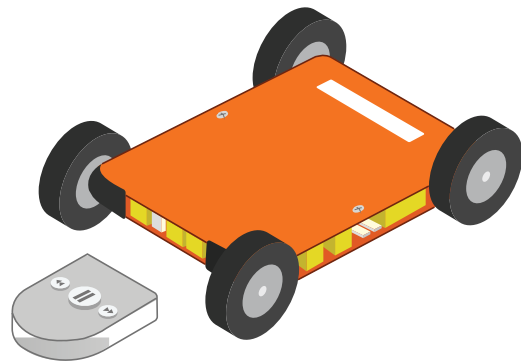


While creating a robotic vehicle, it is important to create a base before adding features. For this, a simple base bot is created.

Base bot

A base bot is a simple robot which understands how to handle tasks based on user inputs. The user can control the movement of the bot using a wired or wireless controller. We can modify the base bot to build more complex bots with multiple sensors for different applications.

In a robotic vehicle, the processor is one of the most important parts. It is also called the 'brain' of the vehicle, along with its controller.



Brain and controller

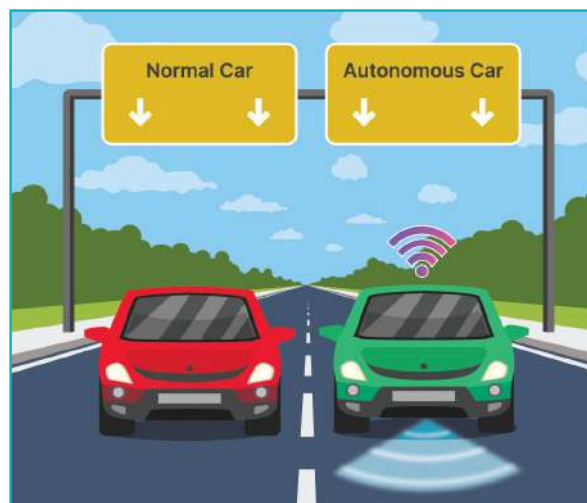
The **Brain**, as the name suggests, is the processing unit of any robot. In any robotic vehicle, the processing unit gets input from sensors and takes decisions based on it. Robots can have a number of ports to connect sensors and actuators.



The **Controller** makes a robot fun and easy to use. With a controller, we can test the robot immediately after it is built. The controller is also called a Joystick. Two analogue joysticks provide fine control to accurately drive and turn a robot, or raise an arm with precise control. Eight buttons provide additional control for various mechanisms that move the robot.

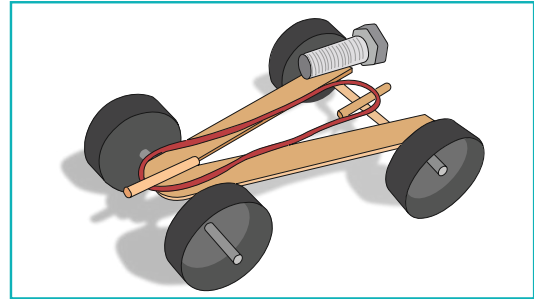


Along with hardware, a robot may also require software, commonly known as a program. Software enables a robot to operate without any human intervention. This type of robot is known as an Autonomous Robot or Automated Robot. A self-driving vehicle is an example of an automated robot.

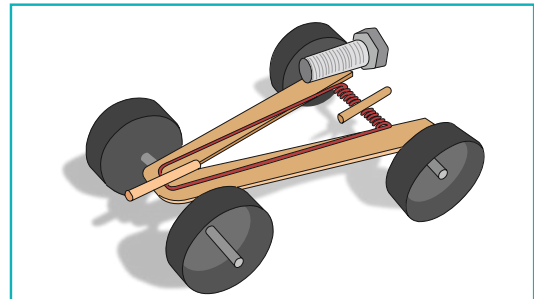


An automated vehicle does not require any input from the user or driver. It gathers information from its surroundings using sensors, processes it, and steers the car by itself.

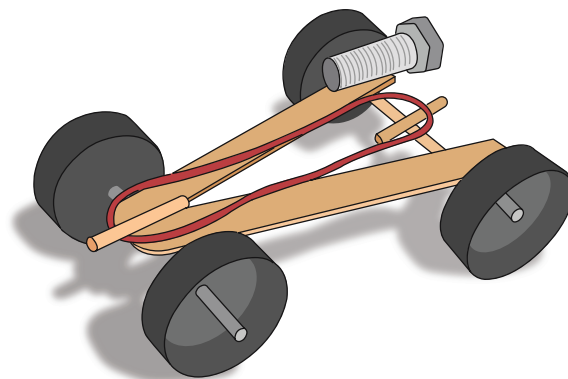
First, let us understand a simple base bot. It can be made without an electronic device, using just a rubber band and ice-cream sticks.



- A basic car structure has 4 wheels and axel. The axel is connected to a rubber band. It can move the bot forward without any external force.
- One end of the rubber band is tied to one pair of wheels. The axel will keep the rubber band fixed. The other end of the rubber band will be wrapped on the other axel.
- It will provide stretched tension that generates the necessary force to move the car forward.



Build a toy car using a rubber band and ice-cream sticks. Try to understand the mechanics behind it.



The above is a basic example of an uncontrolled robotic vehicle. It has a limited distance and can only be used as a toy. To make a controlled and multipurpose bot, hardware kits are available. These can be used to create a simple base bot with a remote controller and processor. VEX IQ Robotic kit is one such pre-moulded hardware kit.

Healthy Practices

1. Use clean protective glasses while working with flying objects or projectiles (parts of the robot that may come loose and fly through the air).
2. Always perform an experiment with approval and under direct supervision of a teacher or another adult.
3. Never leave robotics pieces on the floor. They may cause someone to trip or fall.
4. Do not work with robots around water or with wet hands, since robots and computers work on electricity.

Exercise

Fill in the blanks

- 1 Robotics integrates _____, _____ and _____ to develop machines called robots.
- 2 A robot senses information using inputs from _____ or a controller, analyses using a _____ and acts using _____ as an output.

State whether True or False

- 3 Robotic vehicles are a special type of machine designed to move on the ground, in the air, under water, and in space autonomously, without a human on board.

▲ _____

- 4 Sensors decide the action to perform based on the user input.

▲ _____

- 5 An automated vehicle does not require any input from the user/driver.

▲ _____

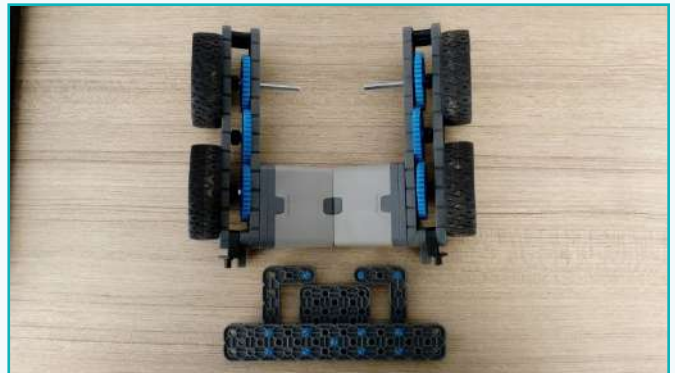
Activity

Build a basic robotic vehicle that can be controlled using a remote controller.

- 1 Gather all the required parts to build a base bot



- 2 Build a structure to connect the motor with the wheel-gear structure. This will provide movement to the bot

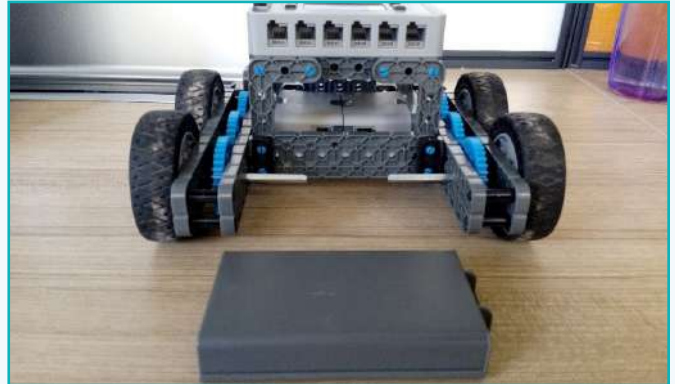


- 3 Connect the motor and the wheel-gear structure to the brain (processor) using connecting wires. This will control the motor based on the user input



Activity

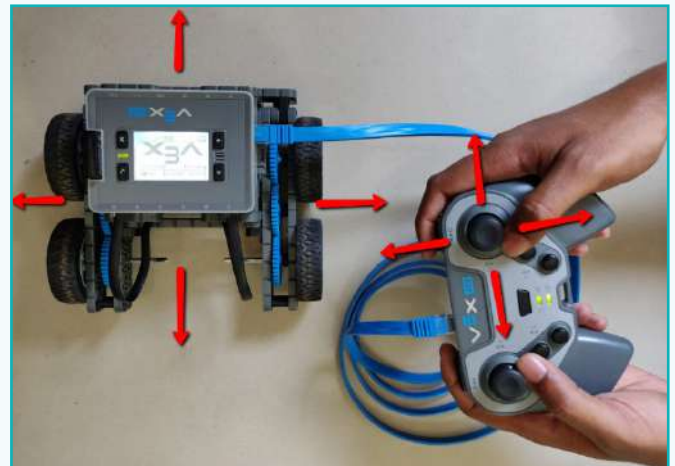
- 4 Connect a power source to provide power to the brain and the motor



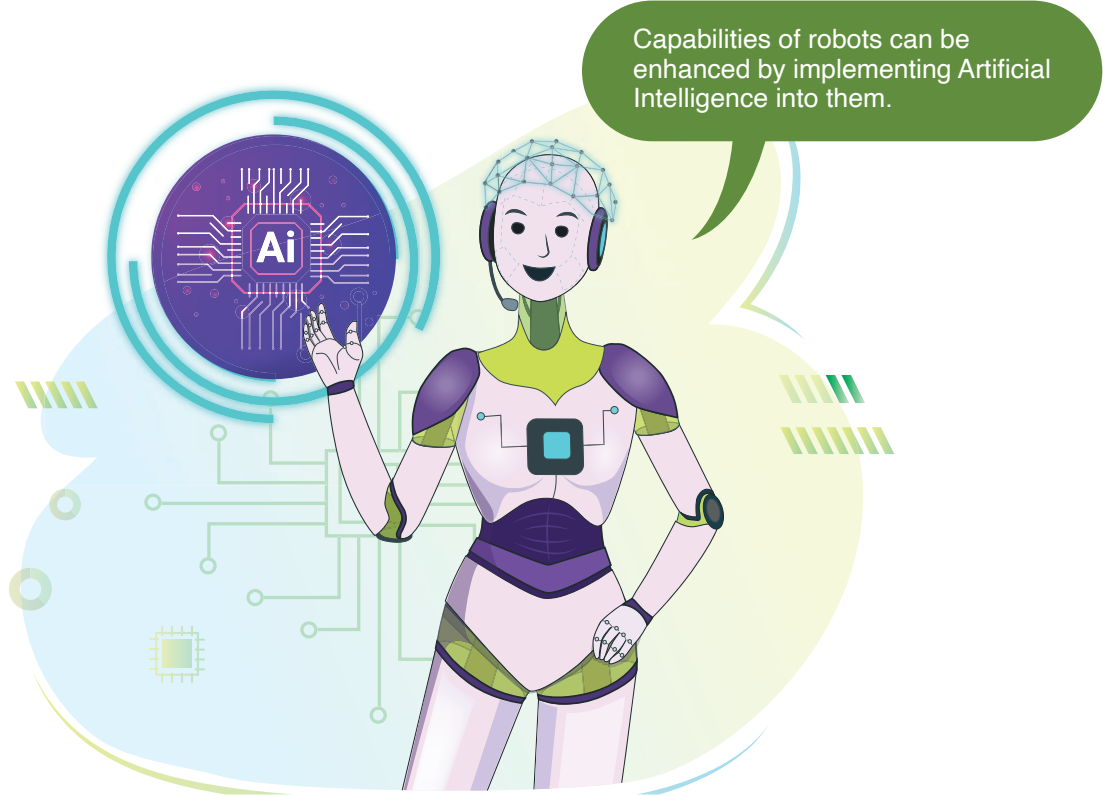
- 5 Connect the controller to provide the user input to the controller. The controller can be connected to the processor using a wire, or even wirelessly



- 6 Switch on the controller and the brain, and move the bot in any direction using a joystick



In this lesson, we learnt about the involvement of robots and robotics in our life. We understood the functions of different parts of a robot, including the brain and controller. We also learnt how to create a base bot.



Tech Flash

- **Robotics** : Robotics is the integration of science, engineering and technology to develop machines called as robots.
- **Robot** : A robot is a machine, particularly one programmable by a computer, capable of automatically carrying out a complex series of actions.
- **Robotic Vehicle** : Robotic vehicles are a special type of machine designed to move on the ground, in the air, under water and in space autonomously, without a human on board.

Future Tech



Human Augmentation

Human Augmentation is a technology that enhances capabilities that a person already has or replaces capabilities that a person has lost. Human augmentations can artificially increase our strength, enhance our sight beyond normal limits, or increase our intelligence too!

